

SARAVANAN P

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building production-ready ML and Generative AI applications using Python, NLP, RAG, and MLOps practices. Strong background in model development, document intelligence, vector search, API deployment, and cloud infrastructure. Built end-to-end AI systems with measurable gains in response quality, retrieval speed, and reliability.

TECHNICAL SKILLS

- **Languages:** Python, SQL
- **AI/ML:** Supervised Learning, Unsupervised Learning, NLP, Computer Vision, Feature Engineering, Model Evaluation
- **GenAI:** LLMs, LangChain, LangGraph, RAG, Prompt Engineering, Ollama, Hugging Face
- **Vector Databases:** Pinecone, FAISS, ChromaDB
- **Deployment:** Flask, FastAPI, REST APIs, Docker (Basics)
- **Cloud/Tools:** AWS (EC2, S3), Git, VS Code, Azure Machine Learning Studio

EXPERIENCE

AI/ML Engineer Intern | ExcelR Solutions, Bangalore | Jun 2025 - Dec 2025

- Completed an intensive AI/ML training program with implementation across classification, regression, clustering, NLP, and deployment use cases.
- Developed and evaluated 5+ ML models using real-world datasets and delivered actionable performance insights.
- Built complete ML workflows including preprocessing, feature engineering, model tuning, validation, and API deployment.
- Improved project delivery quality through structured experimentation and reproducible development practices.

PROJECTS

1) AI Medical Chatbot (LLM + RAG Pipeline) | Dec 2025 - Feb 2026

- Built a medical Q&A assistant using LangChain, Ollama, and RAG with 10,000+ indexed medical documents.
- Improved answer accuracy by 30% and reduced response latency by 20% through retrieval and prompt optimization.
- Implemented Flask/FastAPI services, AWS EC2 deployment, and S3-backed document storage.
- Applied retrieval-grounded response logic to reduce hallucinations and improve answer trustworthiness.
- Tech: Python, LangChain, Ollama, Pinecone, FAISS, Flask, FastAPI, AWS EC2/S3, Docker

2) Multi-Agent Document Processing System (LangGraph) | 2026

- Designed and implemented a multi-agent pipeline for PDF processing: Orchestrator, OCR, Classifier, Metadata, Validation, Extraction, Summary, and RAG Q&A; agents.
- Built structured extraction for contracts, invoices, resumes, and forms with JSON outputs using Pydantic schemas.
- Integrated ChromaDB + sentence-transformers for document-grounded question answering and retrieval.
- Added validation guardrails to improve factual consistency, reduce hallucination risk, and increase output reliability.
- Tech: Python, LangGraph, Gemini 2.5 Flash, PyMuPDF, ChromaDB, sentence-transformers, Streamlit, Flask-ready architecture

EDUCATION

B.Tech, Artificial Intelligence and Data Science | Sri Shakthi Institute of Engineering and Technology, Coimbatore | 2021 - 2025

Relevant Coursework: Machine Learning, NLP, Computer Vision, Data Structures and Algorithms

CERTIFICATIONS

- AI/ML Program, ExcelR Solutions (2025)
- Hugging Face NLP Course (iNeuron)

ADDITIONAL INFORMATION

- GitHub portfolio includes AI applications in LLMs, document intelligence, and computer vision.
- Languages: Tamil (Native), English (Fluent), Kannada (Conversational)